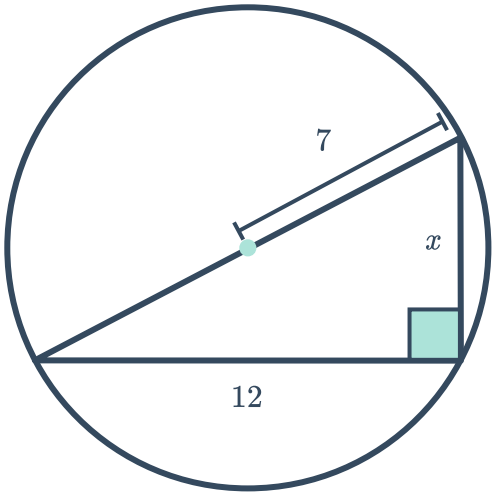
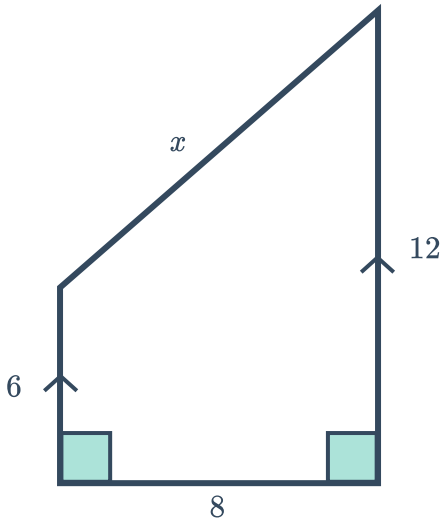
Further applications of trigonometry

1 Find the value of the variable(s) in the following diagrams. Round your answers to two decimal places.

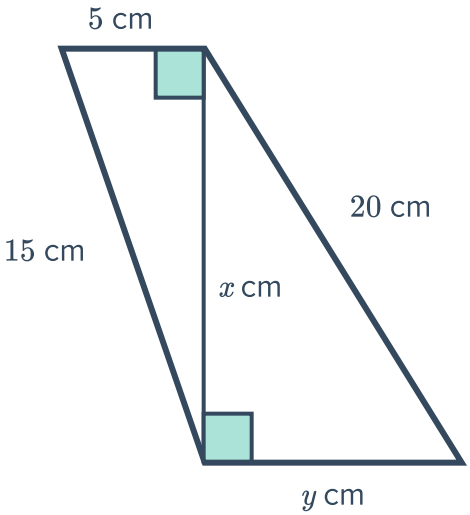
a



b



c



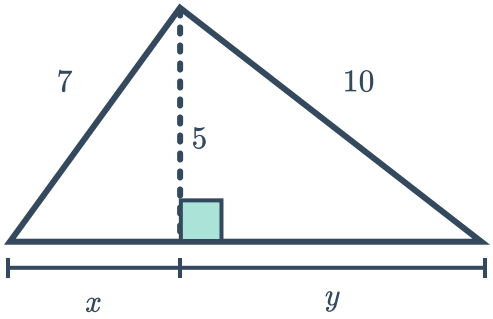
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2 Consider the following diagram.

a Find the value of x, correct to two decimal places.

b Find the value of y, correct to two decimal places.

c Hence, find the length of the base of the triangle to two decimal places.

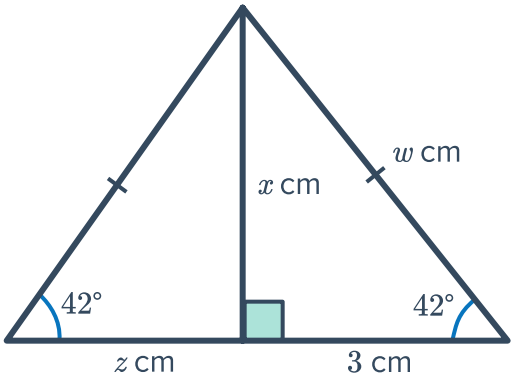


3 Consider the following diagram.

a Find x, correct to two decimal places.

b Find w, correct to one decimal place.

c Find z, correct to the nearest whole number.



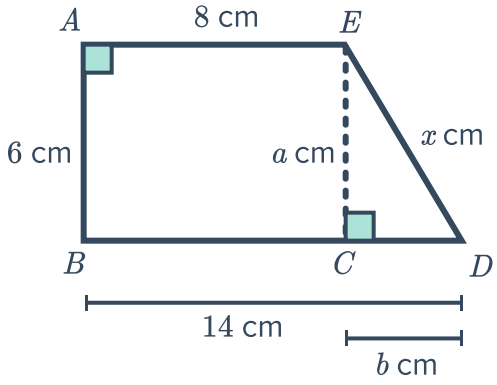
4 Consider the following trapezoid.

a Find the value of a.

b Find the value of b.

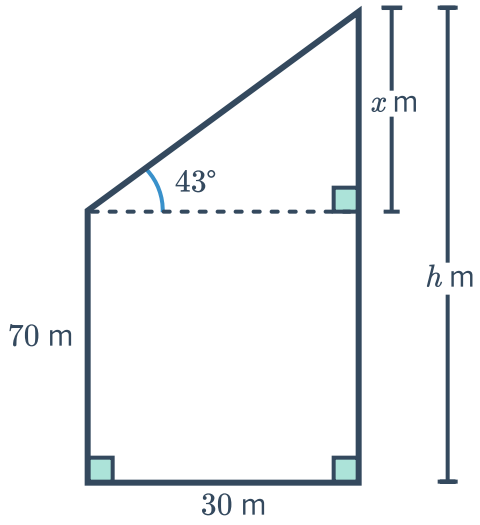
c Find the value of x, correct to two decimal places.

d Hence, find the perimeter of the trapezoid, correct to two decimal places.

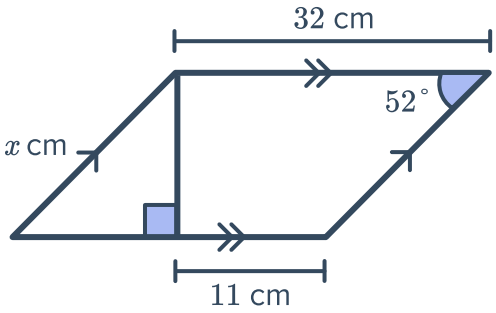


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5 Find the value of h in the given diagram, correct to the nearest integer.



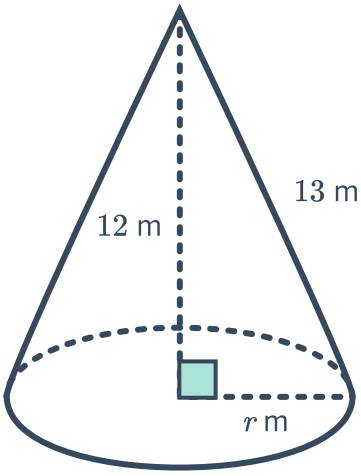
6 Find the value of x, the side length of the given parallelogram, to the nearest centimeter.



7 Consider the cone with slant height of $13\ \text{m}$ and perpendicular height of $12\ \text{m}$.

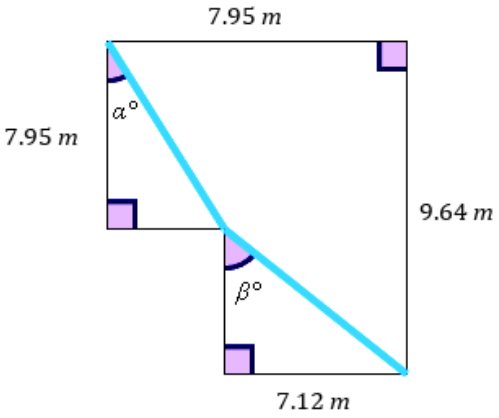
a Find the length of the radius, r.

b Hence, find the length of the diameter of the cone's base.



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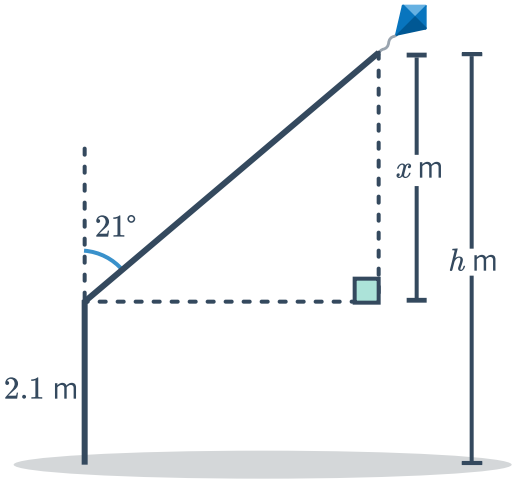
8 The diagram shows the shape and dimensions of Valentina's backyard. Assume that she is going to build a garden path as pictured.



a Find the value of angle α, correct to two decimal places.

b Find the value of angle β, correct to two decimal places.

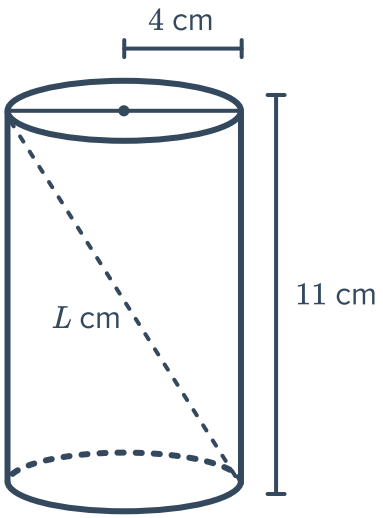
9 A girl is flying a kite that is attached to the end of a 23.4 m length of string. The angle between the string and the vertical is 21°. The girl is holding the string 2.1 m above the ground.



a Find x, correct to two decimal places.

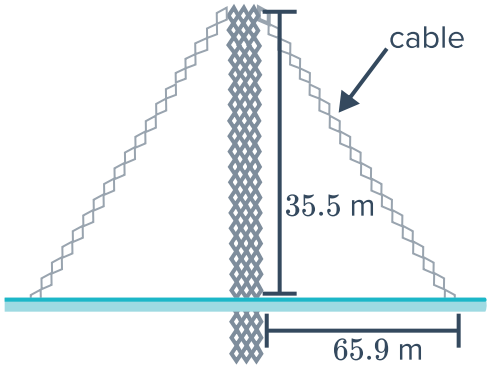
b Hence, find the height, h, of the kite above the ground, correct to two decimal places.

10 A soft drink can has a height of 11 cm and a radius of 4 cm.
Find L, the length of the longest straw that can fit into the can. Round your answer down to the nearest centimeter, to ensure the straw fits inside the can.



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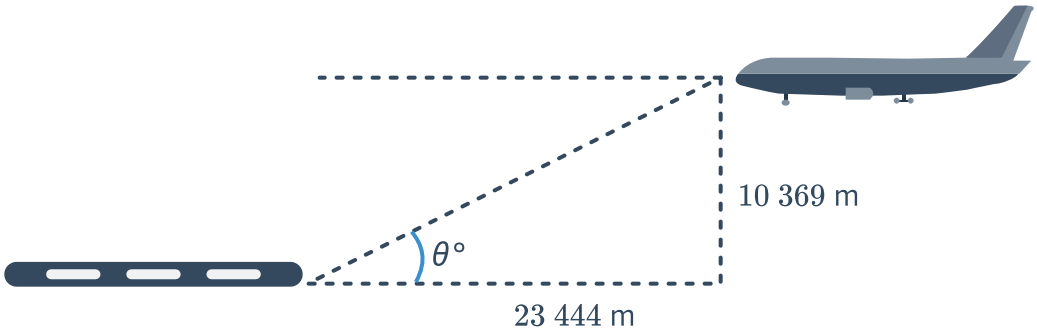
11 A suspension bridge is being built. The top of the concrete tower is $35.5\ \text{m}$ above the bridge and the connection point for the main cable is $65.9\ \text{m}$ from the tower. Assume that the concrete tower and the bridge are perpendicular to each other.



a Find the length of the cable, correct to two decimal places.

b Find the angle the cable makes with the road, correct to two decimal places.

12 The airtraffic controller is communicating with a plane in flight approaching an airport for landing. The plane is $10\,369\ \text{m}$ above the ground and is still $23\,444\ \text{m}$ from the runway. If θ° is the angle at which the plane should approach, find θ, correct to one decimal place.



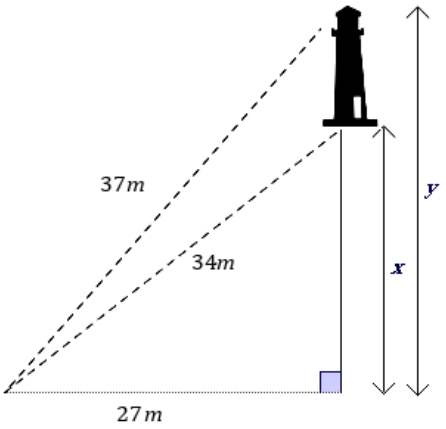
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13 A ship is $27\ \text{m}$ away from the bottom of a cliff. A lighthouse is located at the top of the cliff. The ship's distance is $34\ \text{m}$ from the bottom of the lighthouse and $37\ \text{m}$ from the top of the lighthouse.

a Find the distance from the bottom of the cliff to the top of the lighthouse, y, correct to two decimal places.

b Find the distance from the bottom of the cliff to the bottom of the lighthouse, x, correct to two decimal places.

c Hence, find the height of the lighthouse to the nearest tenth of a meter.

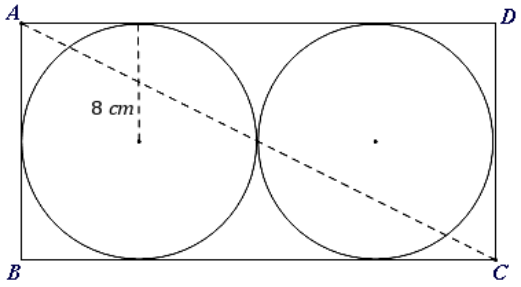


14 Sally is making a rectangular frame to fit two of her favorite clocks side-by-side as shown in the diagram. Each clock has a radius of $8\ \text{cm}$. She wants to put in a diagonal support across the frame.

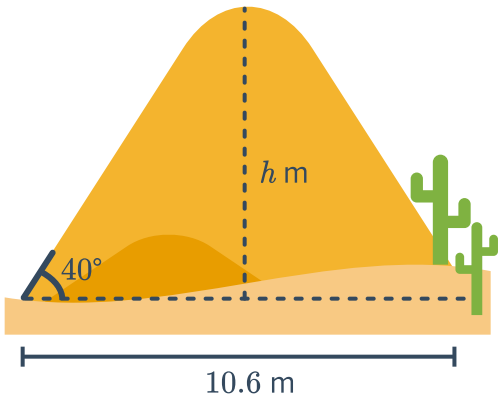
a Find the height of the frame.

b Find the width of the frame.

c Find the length of the diagonal across the frame, correct to two decimal places.



15 Sand has an angle of friction of about 40°. This means once it gets over this it slides down the side of the sand pile.
Mae is getting a delivery of sand to her home, but the area where the sand will be delivered is only $10.6\ \text{m}$ wide.
If h meters is the maximum height of the sand pile, find h, correct to one decimal place.



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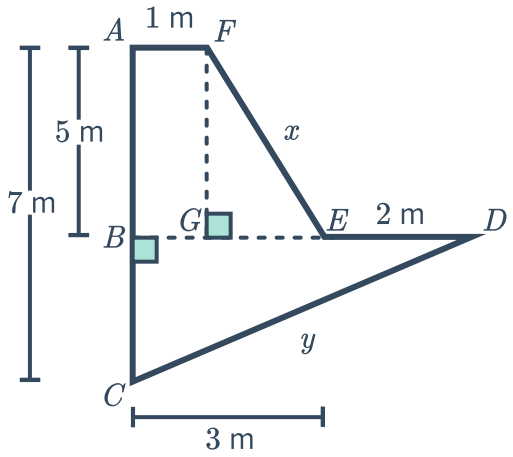
16 A farmer wants to build a fence around the entire perimeter of his land, as shown in the diagram below. The fencing costs $\$37$ per meter.

a Find the value of x, correct to two decimal places.

b Find the value of y, correct to two decimal places.

c How many meters of fencing does the farmer require if fencing is sold by the meter?

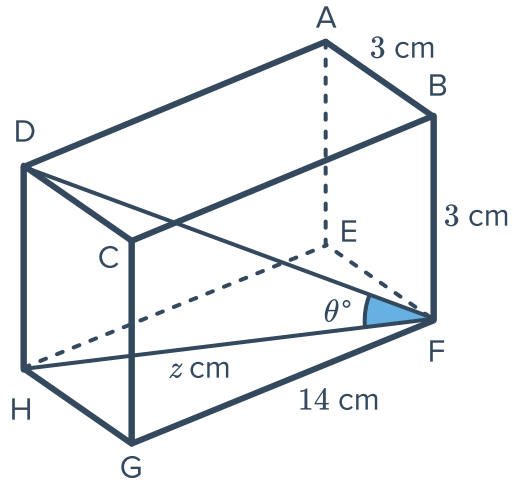
d At $\$37$ per meter of fencing, how much will it cost him to build the fence along the entire perimeter of the land?



17 A square prism has sides of length 3 cm, 3 cm, and 14 cm as shown in the diagram.

a If the diagonal $\overline{HF}$ has a length of z cm, find the value of z, correct to two decimal places.

b If the size of $\angle DFH$ is θ°, find θ, correct to two decimal places.



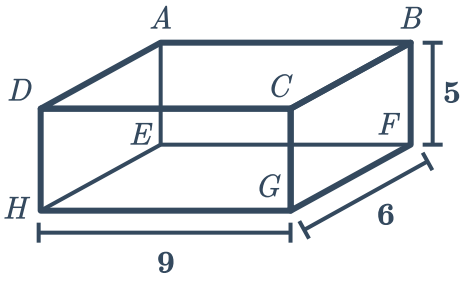
18 A rectangular prism has dimensions as shown in the diagram.

a Find the following lengths in radical form:

i EG ii AG iii DG

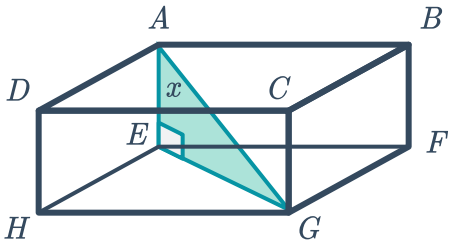
b Find the area of $\triangle ADG$, in surd form.

c Find the size of $\angle DAG$, correct to two decimal places.

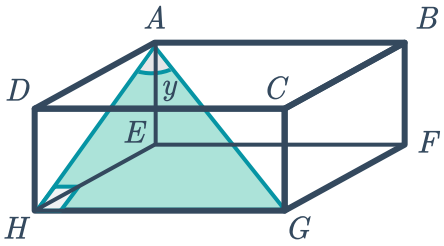


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d A triangular wedge has been inserted into the box as shown.
Find the size of angle x, correct to two decimal places.

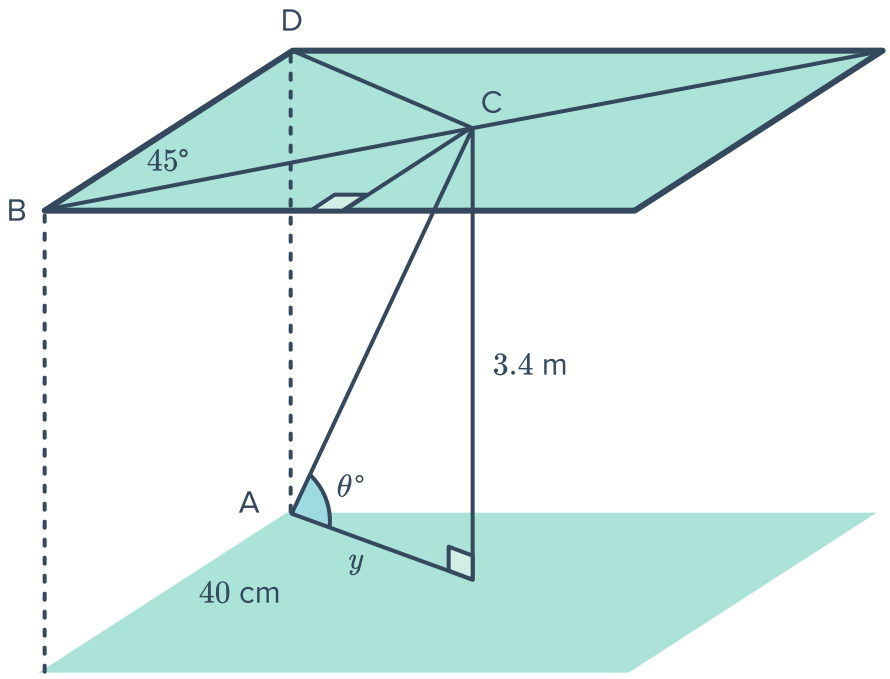


e A different triangular wedge has been inserted into the box as shown.
Find the size of angle y, correct to two decimal places.



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19 A cockroach starts at B and crawls towards C at $10\ \text{cm/min}$ on a $3.4\ \text{m}$ high ceiling. Sally is standing at A.



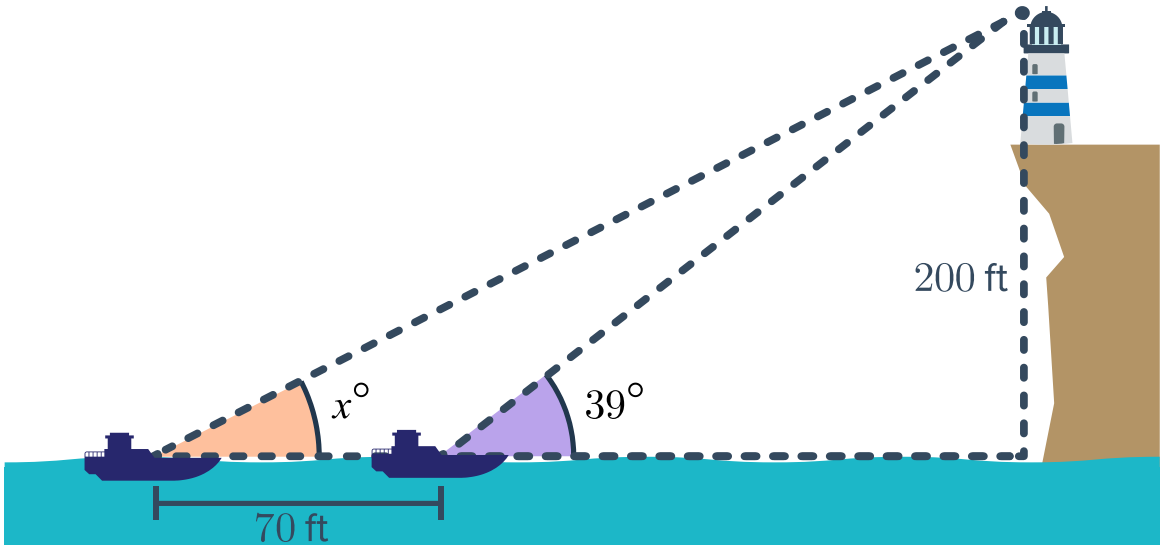
a Calculate the distance, in centimeters, the cockroach will have crawled in the 3 minutes it take to reach C.

b The cockroach is now at C. Calculate y, the distance in centimeters Sally will be from the point directly below the cockroach. Round your answer to two decimal places.

c Calculate θ, the angle of elevation from Sally to the cockroach at C.

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20 Two boats are travelling towards a lighthouse. The top of the lighthouse is $200\ \text{ft}$ above sea level. When the two boats and the lighthouse are collinear, the distance between the captain's windows of each boat is exactly $70\ \text{ft}$. The captain's window of the closest boat has an angle of elevation of 39°.
Find the value of x. Round your answer to the nearest hundredth of a degree.

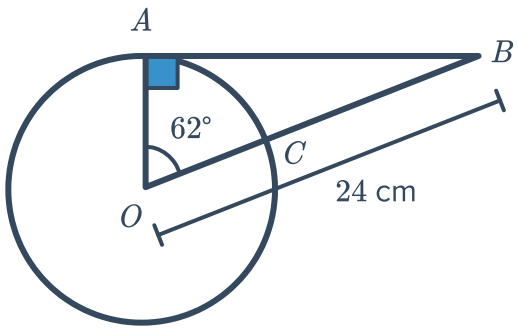


Additional questions

21 AB is a tangent to a circle with centre O.
OB is $24\ \text{cm}$ long and cuts the circle at C.

a Find the length of the radius of the circle, r, rounding your answer to one decimal place.

b Find the length of BC, rounding your answer to one decimal place.



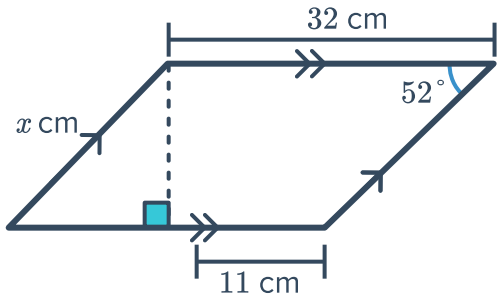
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22

a Find the value of x, the side length of the parallelogram, rounding your answer to two decimal places.

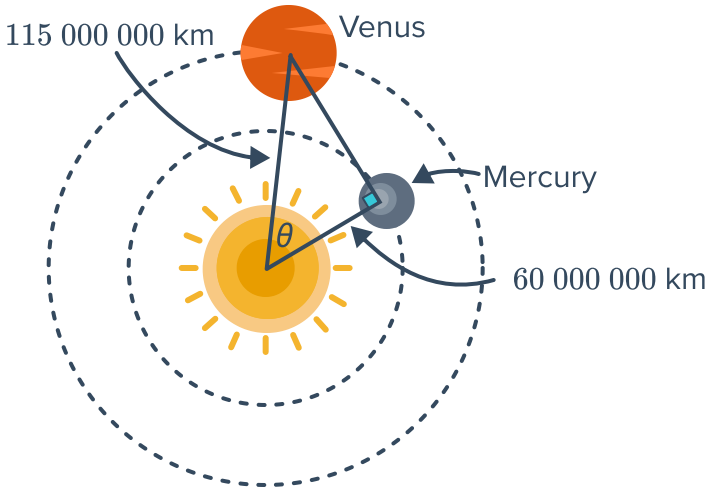
b Describe a method you could use to find the area of the parallelogram.

c Find the area of the parallelogram correct to the nearest square centimeter.



23 An isosceles triangle has a base of length $53\ \text{in}$ and the angle opposite the base has a measure of 31°. What is the area of the triangle?

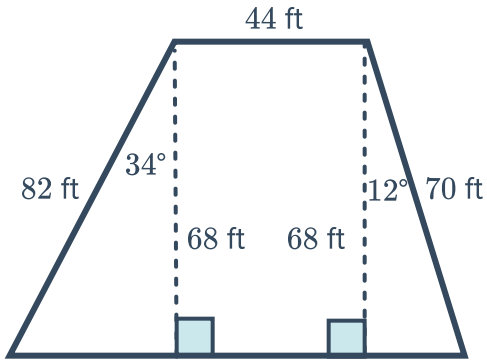
24 During rare parts of Mercury and Venus' orbit, the angle from the Sun to Mercury to Venus is a right angle, as shown in the diagram:



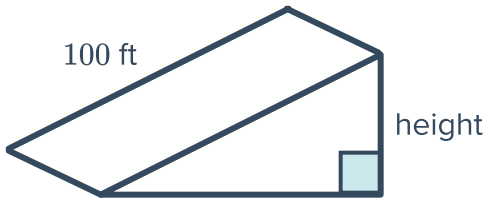
The distance from Mercury to the Sun is $60\,000\,000\ \text{km}$. The distance from Venus to the Sun is $115\,000\,000\ \text{km}$.
What is the angle θ, from Venus to the Sun to Mercury? Round your answer to the nearest tenth.

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25 Latoyah and Thiago's school has a grassed playground in the shape of a trapezoid, that they want to design a shade cloth to put over it. They need to calculate the area and, after measuring various parts, drew the following diagram.
Determine the area of the playground as illustrated in the diagram. Round your answer to the nearest meter.



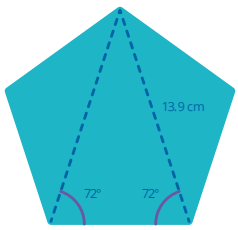
26 A ramp of length 100 ft needs to ascend at an angle between 10° and 15° for it to be safe to use.
Find a suitable height for the ramp to ensure it is safe. Justify your decision.



27 Eboni needs to use a piece of paper in the shape of a regular pentagon to make an origami flower. The first step of the process is to fold along the dotted lines shown. The length of each fold line is 13.9 cm.

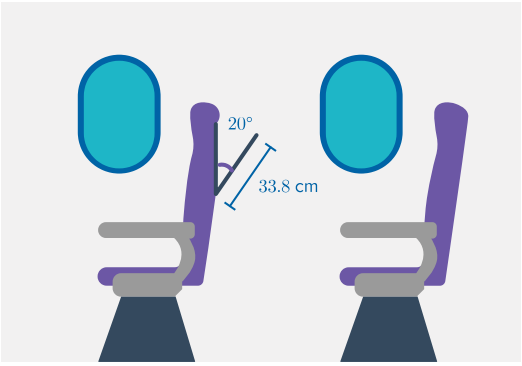
a Find the perpendicular height of the piece of paper. Round your answer to the nearest tenth.

b Find the length of each side of the paper. Round your answer to the nearest tenth.



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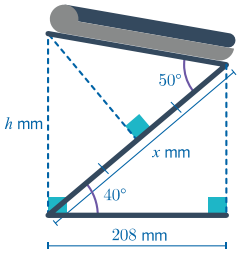
28 On Lerringdu Airlines, each passenger has a screen for entertainment attached to the chair in front of them. This screen itself has a height of $33.8\ \text{cm}$ and is able to tilt away from its frame by 20°.
How far does the screen reach out from the seat in front when it is tilted away by 20°?
Round your answer to two decimal places.



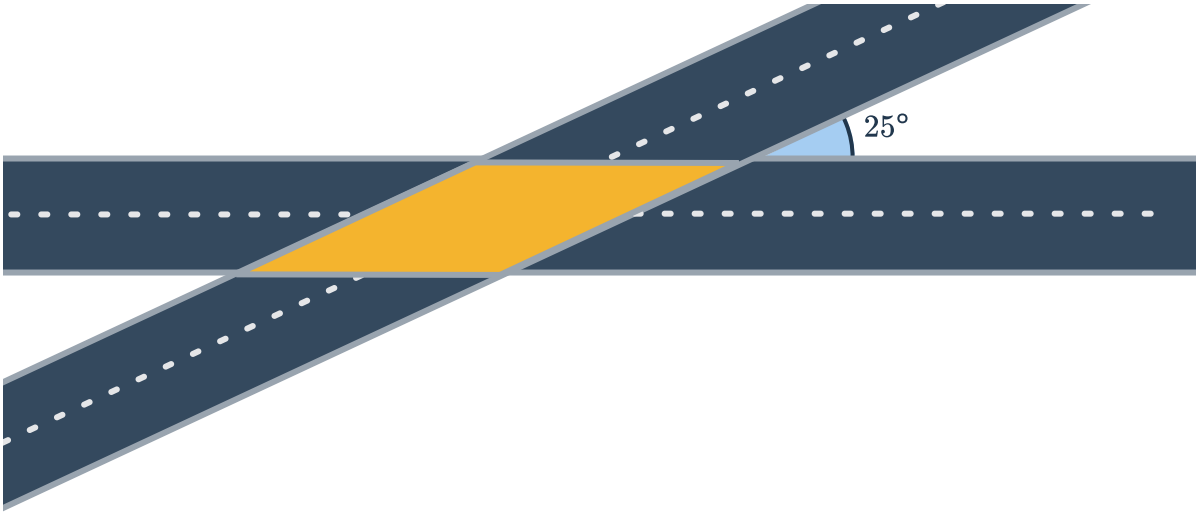
29 Roula is looking to buy a particular stand for her laptop. The picture of the stand in the catalogue shows the maximum angle of each part of the stand, along with the length of the stand's base.
Roula wants to figure out the height of the stand when it is fully opened up.

a First find x, the length of the middle section of the stand. Round your answer to two decimal places.

b Find the value of h, the height of the stand. Round your answer to two decimal places.



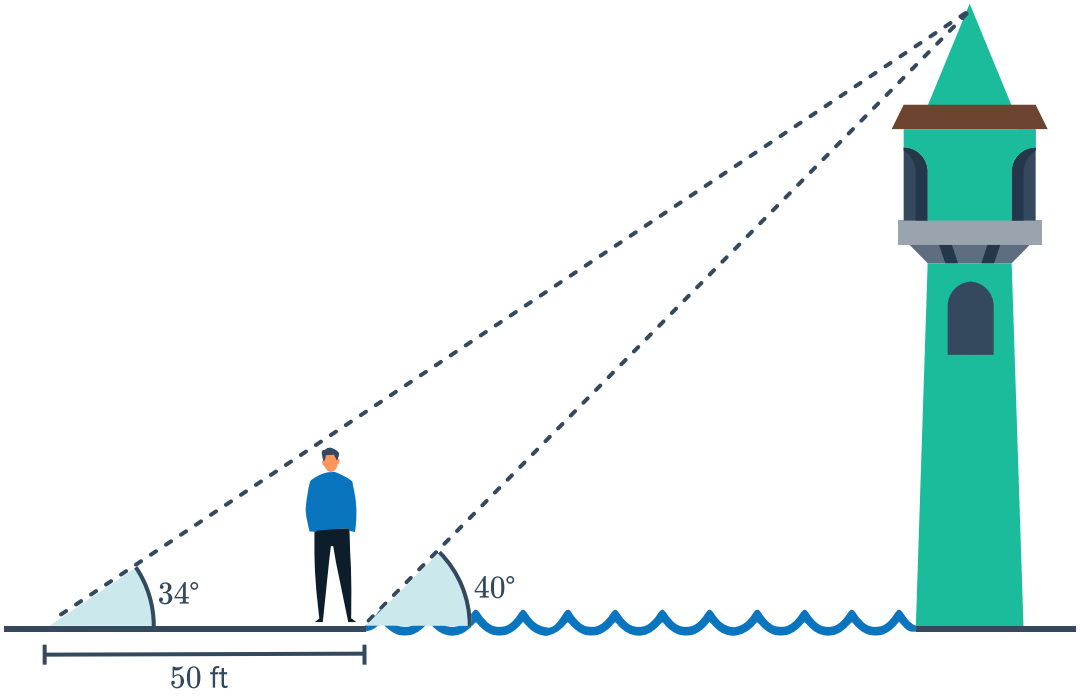
30 Two roads intersect to make an angle with a measure of 25°. Both roads are 20 foot in width.



The intersection forms the yellow parallelogram shown. Find the area of the paralellogram.
Give your answer to the nearest tenth.

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31 Tyrese has been asked to work out the height of the Ponce de Leon Inlet Light. He measures the angle from the beach to the top of the lighthouse to be 34° and then walks 50 feet closer to the lighthouse and measures the angle to be 40°. Set up two equations in terms of the horizontal distance to the tower and the unknown height of the tower, and solve to find the height. Round your answer to the nearest foot.



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